

ANSH RAJ SURYAVANSHI

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Summary

Software Engineer / AI Engineer with hands-on industry experience building production machine learning and LLM-based systems. Specialized in agentic RAG, edge AI, and computer vision, with experience deploying real-time, in-vehicle systems in automotive environments.

Education

Kettering University

Flint, MI

B.S. in Computer Science, Concentration in AI, Minor in Applied Mathematics

October 2021 - September 2025

- **GPA:** 3.94
- **Activities:** Facility Manager at Rec Center, Volleyball Club, VP at International Club
- **Honors:** Dean's List 2021-2025, UPE CS & KME Math Honor Societies, Hack Dearborn 2023/2025 Winner (ZF's Automotive & Amazon's Financial Firewall Tracks), \$10,000 MEDC Michigan Scholar Award

Experience

General Motors

Flint, MI

Software Engineer

March 2026 - Present

- Leading deployment of 72 AI cameras across 38 conveyors in the GM Flint plant, enabling real-time computer vision monitoring at scale.
- Developed an end-to-end automation pipeline for camera configuration, covering onboarding, pilot, and production, reducing setup time from 5-6 weeks to 2-3 days for TPMs across 7 GM plants.
- Maintained and updated technical documentation and tools for the camera onboarding process, ensuring accuracy across pipeline changes.

Hyundai Mobis

Plymouth, MI

AI/ML Engineer (Rotational Co-op, 12 months FTE)

October 2022 - June 2025

- Optimized a local LLM with agentic RAG for in-vehicle edge deployment, reducing inference latency by 35% and enabling a real-time voice-based driver alertness system, improving engagement by 20%.
- Developed an ML-based hand gesture recognition system for in-cabin controls; showcased at CES 2023 and planned for deployment across 5M+ vehicles by 2025.
- Built reusable backend libraries for math, vector, and matrix operations, eliminating 30% of redundant code; added OpenCV, OpenGL, and Vulkan-based frameworks and automated CI/CD pipelines, cutting manual setup by 40%.

Projects

Undergraduate Co-op Thesis

Kettering University & Hyundai Mobis | April 2025 - Present

- Built a local LLM-based reasoning system using agentic RAG and multi-modal sensor data to enable low-latency, on-device in-cabin comfort and entertainment recommendations on a Cruden Simulator.

Taboo Party - Real-Time Multiplayer Word Game

Founder & Developer | December 2025 - Present

- Built and deployed a real-time multiplayer word game integrated as an official Discord Activity leveraging Next.js/React, TypeScript & Socket.IO, reaching 52,000+ visits & 800+ registered users with 100+ server community members.

Givvy - Encrypted Gift Cards

Hack Dearborn 2025 Winner | October 2025

- Built an AES-encrypted NFC e-ink gift card platform using TypeScript and Supabase with web and POS apps, winning Hack Dearborn 2025 among 240+ participants.

Recreation Center App

Personal Project | July 2025 - September 2025

- Built REC-IT, a full-stack React and TypeScript web app using Supabase and Google Firebase to replace manual recreation center workflows.

Gigs for Pi

Personal Project | June 2024 - December 2024

- Built a Web3 freelancing platform using TypeScript and Supabase, reaching 20,000+ likes, a 4.78/5 rating across 37,000+ reviews, and a feature on the official Pi Network GitHub.

Road Entertainment System

Hack Dearborn 2023 Winner | October 2023

- Built an ML-powered in-cabin entertainment recommendation system using trip ETA and user preferences, achieving 89% accuracy and integrating Google Maps API with hand-gesture controls.

Skills

Programming Languages: C, C++, Python, Java, SQL, TypeScript, JavaScript, R, MATLAB

Machine Learning & AI: Machine Learning, Computer Vision, LLMs, Agentic AI, RAG, Model Fine-Tuning, LLM Distillation, PyTorch, TensorFlow, Scikit-Learn, Agent Orchestration, Multimodal AI, AI-Assisted Development

Systems & Backend: Flask, REST APIs, CI/CD, Jenkins, Git, GitHub, Distributed Systems, Edge AI, Agile, SDLC, Jira

Data & Databases: PostgreSQL, MongoDB, Vector Databases (ChromaDB, FAISS), Supabase, Pandas, NumPy

Cloud & Platforms: AWS (S3, Lambda, Amplify), Google Cloud (GCP), Docker, Oracle, Google Firebase

Tools & Frameworks: React, Next.js, Socket.IO, LangChain, LangGraph, Agno, OpenCV, OpenGL, Vulkan, Streamlit